

1. For the network shown in Fig. 1, $R_1 = 600\text{K}\Omega$, $R_2 = 400\text{K}\Omega$ and $C = 0.2\mu\text{F}$. Determine the maximum phase provided by this network and the frequency (in Hz) at which this phase occurs. [6]

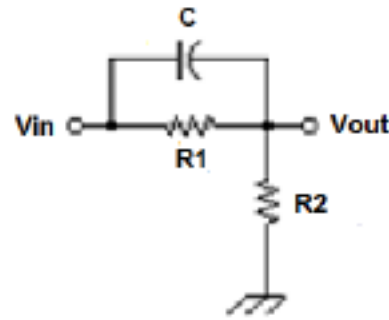


Fig.1

Solution:

T.F is $G(s) = \frac{V_{out}(s)}{V_{in}(s)} = \frac{Z_2(s)}{Z_1(s) + Z_2(s)}$ where $Z_1(s) = R_1$ in parallel $1/Cs$ and $Z_2(s) = R_2$

$$Z_1(s) = \frac{R_1}{R_1Cs + 1} \quad \text{Substituting the values of } Z_1(s) \text{ and } Z_2(s), G(s) = \frac{R_2(R_1Cs + 1)}{R_2R_1Cs + R_1 + R_2}$$

$$\text{Dividing by } R_2, G(s) = \frac{R_1Cs + 1}{R_1Cs + \frac{R_1 + R_2}{R_2}}$$

$$\text{Substituting the values, } G(s) = \frac{0.12s + 1}{0.12s + 2.5}$$

$$\text{Phase of } G(s) \text{ is } \phi = \tan^{-1}(0.12\omega) - \tan^{-1}(0.048\omega) \text{ ----(1)}$$

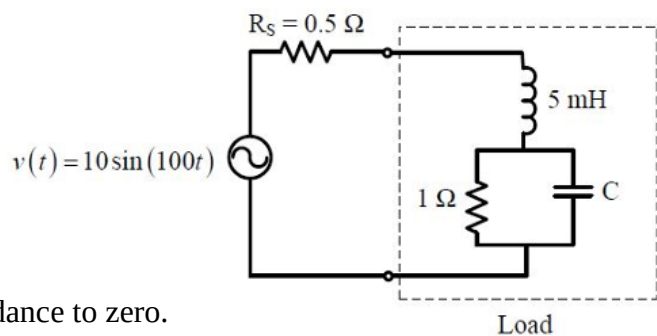
$$\Phi \text{ will be maximum when } \tan\Phi \text{ is maximum. } \tan\phi = \frac{0.12\omega - 0.048\omega}{1 + (0.12 \times 0.048)\omega^2}$$

$$\text{Differentiating and equating to zero, we get } \omega = \frac{1}{\sqrt{0.12 \times 0.048}} = 13.176 \text{ rad/sec}$$

$$\text{Frequency in Hz} = 2.097 \text{ Hz}$$

$$\text{Substituting in (1), } \phi = 25.3769 \text{ degrees.}$$

2. In the circuit shown in Fig. 2, determine the value of the capacitor required for maximum power to be transferred to the load.



Solution:

Equate the imaginary part of the load impedance to zero.

$$Z_L = j\omega L + \frac{R \times \frac{1}{j\omega C}}{R + \frac{1}{j\omega C}} = j\omega L + \frac{R}{1 + j\omega RC} = j\omega L + \frac{R(1 - j\omega RC)}{1 + \omega^2 R^2 C^2}$$

Equating the j term equal to zero

$$\Rightarrow \omega L = \frac{\omega R^2 C}{1 + \omega^2 R^2 C^2} \Rightarrow L = \frac{R^2 C}{1 + \omega^2 R^2 C^2} \Rightarrow 5 \times 10^{-3} = \frac{C}{1 + 10^4 C^2}$$

$$\text{Solving } C = 10 \text{ mF.}$$

3. The network $\mathcal{N}$ shown in Fig. 3a and 3b is linear, bilateral and passive. Determine the current I in Fig. 3b. [7]

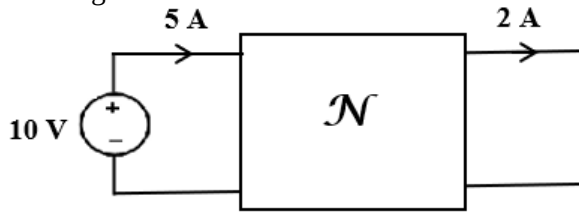


Fig. 3a

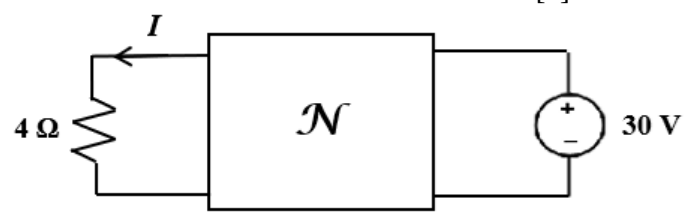
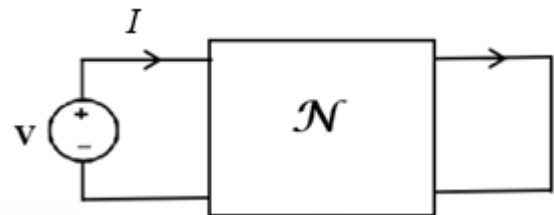


Fig. 3b

Solution:

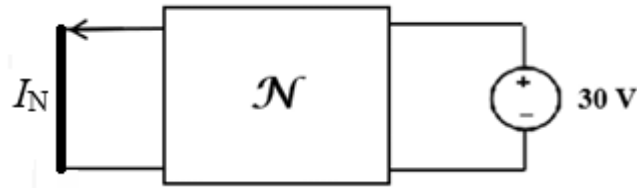
You can find the current through 4Ω resistance using the Norton's Theorem.

To find the Norton's resistance R_N , remove the 4Ω and apply a voltage and short circuit the 30 V voltage source as shown in figure below. $R_N = \frac{V}{I}$

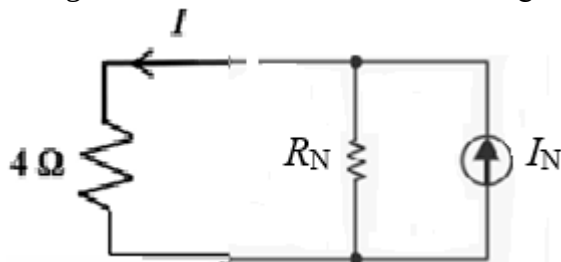


If we apply 10 V , the current I is 5 A $R_N = \frac{10}{5} = 2\Omega$ (Fig. 3a)

To find the Norton's current I_N , short the 4Ω resistance as shown in the circuit below. and find the short circuit current.



Since 10 V produces 2 A response (See the first figure in question 3 a), 30 V produces 6 A in the above figure. $I_N = 6\text{ A}$. So the current through 4Ω resistance $I = 2\text{ A}$



$$I = 6 \times \frac{2}{4+2} = 2\text{ A}$$

4. The switch in the circuit in Fig. 4 has been open a long time before closing at $t=0$. Find $i_L(t)$ for $t \geq 0$. [9]

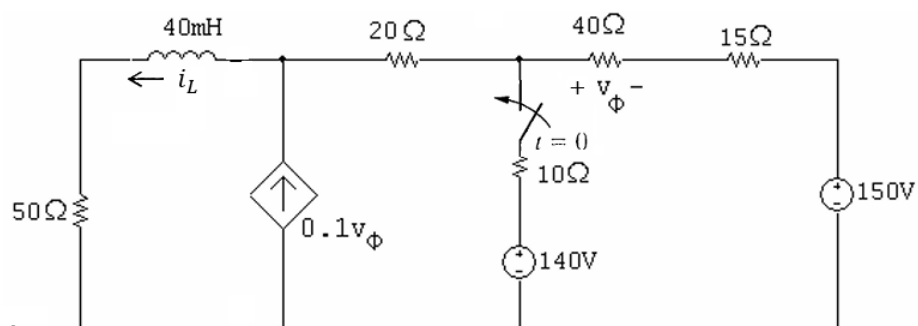
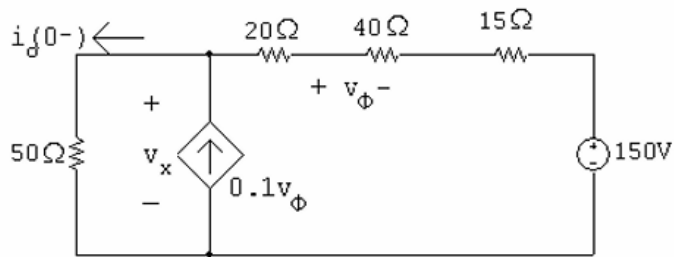


Fig. 4

Solution:

For $t < 0$



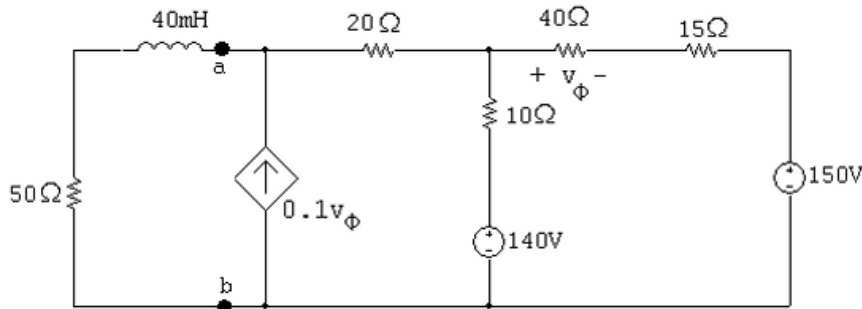
$$\frac{v_x}{50} - 0.1v_\phi + \frac{v_x - 150}{75} = 0$$

$$v_\phi = \frac{40}{75}(v_x - 150)$$

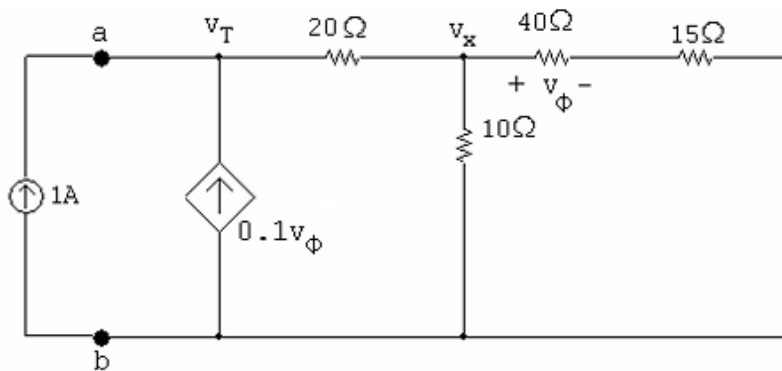
Solving, $v_x = 300 \text{ V}$;

$$i_o(0) = \frac{v_x}{50} = 6 \text{ A}$$

$t > 0$



Find the Thevenin's equivalent resistance with respect to a,b. Apply a 1 A source



$$-1 - 0.1v_\phi + \frac{v_T - v_x}{20} = 0$$

$$\frac{v_x - v_T}{20} + \frac{v_x}{10} + \frac{v_x}{55} = 0$$

$$v_\phi = \frac{40}{55}v_x$$

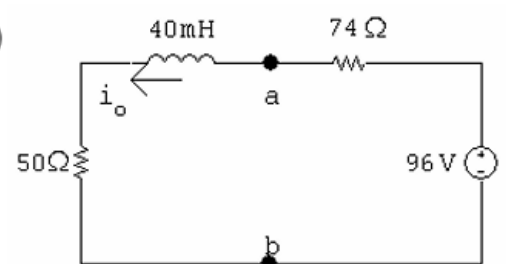
Solving, $v_T = 74 \text{ V}$ so $R_{Th} = \frac{v_T}{1 \text{ A}} = 74 \Omega$

Find the open circuit voltage with respect to a, b:

$$-0.1v_\phi + \frac{v_{Th} - v_x}{20} = 0$$

$$\frac{v_x - v_{Th}}{20} + \frac{v_x - 140}{10} + \frac{v_x - 150}{55} = 0 \quad v_\phi = \frac{40}{55}(v_x - 150)$$

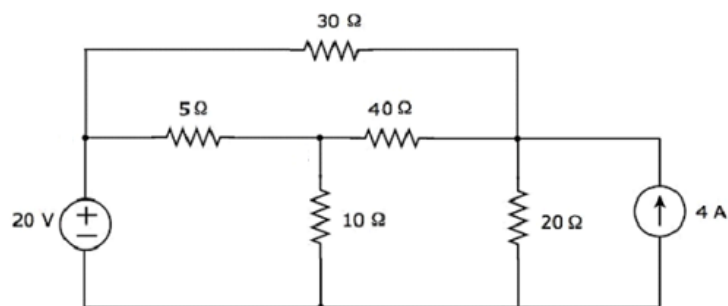
Solving, $v_{Th} = 96 \text{ V}$



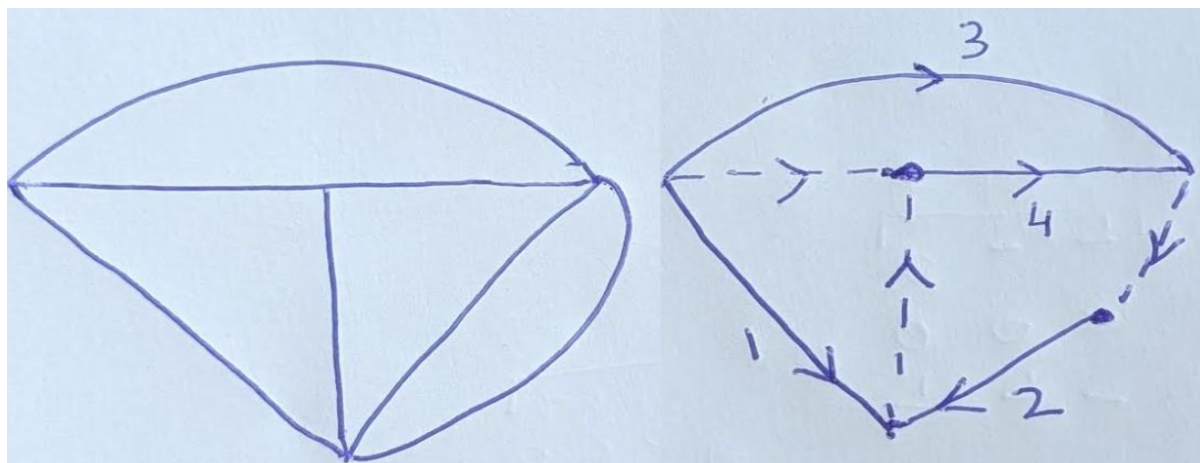
$$i_o(\infty) = 96/124 = 0.774 \text{ A} \quad \tau = \frac{40 \times 10^{-3}}{124} = 0.3226 \text{ ms}; \quad 1/\tau = 3100$$

$$i_o = 0.774 + (6 - 0.774)e^{-3100t} = 0.774 + 5.226e^{-3100t} \text{ A}, \quad t \geq 0$$

5. Draw appropriate graph and obtain the topological formulation of loop equations for the network shown in Fig.5. Follow the convention studied in the class for drawing the graph and the formulation of loop equations. Take $30\ \Omega$ and $40\ \Omega$ branches as tree branches. [7]



Solution:



$$\begin{array}{c}
 \begin{matrix} e_1 & e_2 & e_3 & e_4 & e_5 & e_6 & e_7 \\
 5 & \begin{bmatrix} -1 & 1 & 1 & 0 & 1 & 0 & 0 \end{bmatrix} \\
 6 & \begin{bmatrix} 1 & 0 & -1 & 1 & 0 & 1 & 0 \end{bmatrix} \\
 7 & \begin{bmatrix} 0 & 0 & -1 & 1 & 0 & 0 & 1 \end{bmatrix} \\
 B_E & B
 \end{matrix}
 \end{array}
 \quad
 \begin{array}{c}
 E_b = \begin{bmatrix} E_1 \\ E_2 \end{bmatrix} \begin{matrix} 20 \\ 80 \end{matrix} \\
 I_c = \begin{bmatrix} I_5 \\ I_6 \\ I_7 \end{bmatrix}
 \end{array}$$

$$\underline{B Z_b B^T I_c = -B_E E_b}$$

$$Z_b = \begin{bmatrix} 30 & 0 & 0 & 0 & 0 \\ 0 & 40 & 0 & 0 & 0 \\ 0 & 0 & 20 & 0 & 0 \\ 0 & 0 & 0 & 10 & 0 \\ 0 & 0 & 0 & 0 & 5 \end{bmatrix}
 \quad
 -B_E E_b = \begin{bmatrix} E_2 - E_1 \\ E_1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 50 & -30 & -30 \\ -30 & 80 & 70 \\ -30 & 70 & 75 \end{bmatrix}
 \begin{bmatrix} I_5 \\ I_6 \\ I_7 \end{bmatrix} = \begin{bmatrix} -60 \\ -20 \\ 0 \end{bmatrix}$$

$$Z I_c = E \quad I_c = \ln v(Z) E = \begin{bmatrix} 7.6627 \\ -1.5901 \\ 0.8191 \end{bmatrix}$$

6. In the circuit shown in Fig. 6, the input is $I(t)$ and the output is $V(t)$.
- Obtain the state variable model of the circuit.
 - Find a relationship between R , L , and C that renders the system uncontrollable.
 - Determine the network function relating $V(s)$ to $I(s)$ when the system is uncontrollable.
 - Is the system observable when it is uncontrollable? [8]

Solution:

Select capacitor voltage as x_1 (with polarity shown) and inductor current as x_2 (with direction shown).

Writing KCL at capacitor junction, $C\dot{x}_1 + x_2 - I(t) = 0$

$$\dot{x}_1 = \frac{-1}{C}x_2 + \frac{1}{C}I(t) \dots\dots\dots (1)$$

Writing KVL in the inductor loop, $L\dot{x}_2 + Rx_2 - x_1 + R(x_2 - I(t)) = 0$

$$\dot{x}_2 = \frac{1}{L}x_1 - \frac{2R}{L}x_2 + \frac{R}{L}I(t) \dots\dots\dots (2)$$

Output equation is $V(t) = x_1 - Rx_2 + RI \dots\dots\dots (3)$

From the above equations [2 Marks]

$$A = \begin{bmatrix} 0 & -\frac{1}{C} \\ \frac{1}{L} & -\frac{2R}{L} \end{bmatrix} \quad B = \begin{bmatrix} \frac{1}{C} \\ \frac{R}{L} \end{bmatrix} \quad \text{and} \quad C = [1 \quad -R]$$

Controllability matrix $\begin{bmatrix} 1/C & -R/LC \\ R/L & (L - 2R^2C)/L^2C \end{bmatrix}$

System is uncontrollable when the above matrix loses rank. Setting the determinant equals to zero

$$L = R^2C \quad [2 \text{ Marks}]$$

Transfer function is $Z(s)$

$$Z(s) = \left(R + \frac{1}{Cs} \right) \text{ is in parallel with } R + Ls = \frac{RLCs^2 + (L + RC^2)s + R}{LCs^2 + 2RCs + 1}$$

Putting $L = R^2C$ $Z(s) = R$ [2 Marks]

Observability matrix $\begin{bmatrix} 1 & -R \\ -R/L & (2R^2C - L)/LC \end{bmatrix}$ Putting $L = R^2C$, the determinant becomes zero.

The system is unobservable when it is uncontrollable [2 Marks]